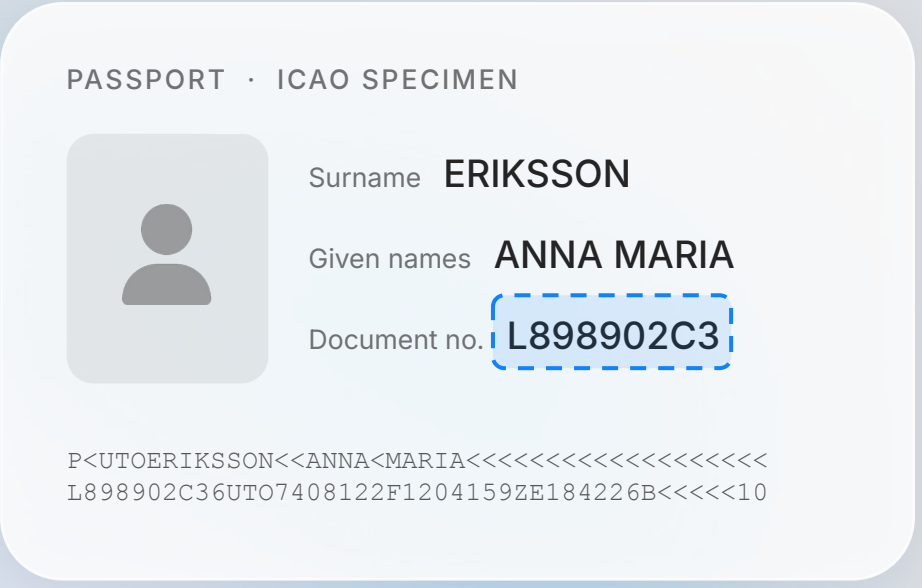
ITAI 1378 Computer Vision and AI · Midterm Project: The Blueprint

Document Autopilot

From a phone photo of a passport to verified data, confirmed by a person in one click.



Project tier

Tier 3

Solo · 16 weeks

Jiri Musil

Built for recruitment agencies' visa teams

The Problem

Every passport is manually retyped, with no checks to catch typing errors.

Who cares

Agency visa officers, and workers waiting to travel

Today

Blurry phone photos, every value typed by hand

Why it matters

One wrong digit can send a visa application back

The fix

Read it, prove it, let a person confirm it

= Plain OCR today

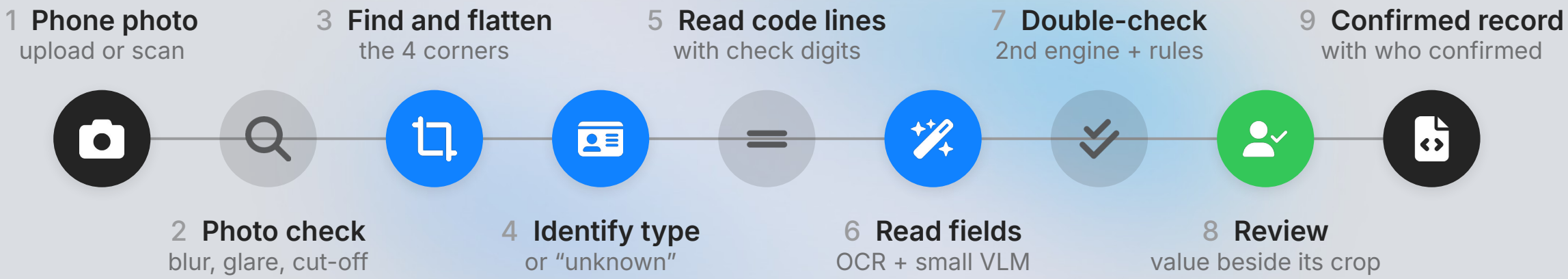
10.5%

of passport code lines read fully right

Tesseract 4 on the MIDV-2020 benchmark

Solution Workflow

A phone photo turns into verified data. Nothing is saved until someone confirms it.



Machine rules · AI proposes · Person decides · In and out

Technical Approach

Five open-source models work in sequence. Exact readers go first. AI steps in only to fill the gaps.

STEP	CV TECHNIQUE	ARCHITECTURE	MODEL	HOW I USE IT
Find the page	Detection	CNN	YOLO11n-pose	Transfer learning
Name the document	Classification	CNN	EfficientNet-B0	Transfer learning
Read the text	OCR	CNN + Transformer	PP-OCRv5	Pretrained
Fill the gaps	VLM	ViT + language model	PaddleOCR-VL 0.9B	Pretrained
Double-check	OCR re-read	Vision Transformer	docTR	Pretrained

Frameworks PyTorch | Hugging Face | PaddleOCR | OpenCV | Gradio *all free and open source*

Why double-check? Check digits caught every incorrect digit, but missed 8% of wrong letters.

Data Plan

A free public set of mock IDs, with labels included. No real person's documents.

MIDV-2020 · public, free

1,000 mock IDs	10 document types
72,409 labelled images	0 real people

arxiv.org/abs/2107.00396

≈ 13,000 images
Every photo and scan + 10 frames per video

Labels already there
Type, 4 corners, every field's value and box

Fair split, by document
700 train · 150 validation · 150 test

Extra tests
5,000 synthetic code lines + ICAO specimens

Success Metrics: My Promise

Measured on 150 documents the system has never seen.

Primary · accuracy

≥ 95%

of critical fields read exactly right

plain OCR today: 10–39%

Secondary · speed

≤ 5 s

per document, on a free Colab GPU

photo in → review screen ready

Must always hold

- Check digits always right
- No value without proof
- Unknown types flagged

Milestone Plan

Something runs end to end before any training.

Week 10

Blueprint

Plan, repo, first notebook

Midterm submitted

Week 11

First Working Demo

Pretrained OCR on 10 photos

Something works

Weeks 12–13

Make It Yours

Fine-tune, add the VLM, build the review screen

Works on ID documents

Week 14

Improve and Measure

Test on 150 unseen documents

Metrics recorded

Week 15

Package and Present

Demo video, README, slides

Final Work submitted

Risks and Resources

Two real risks, each with a Plan B. Everything runs free.

Dataset access

Medium

Colab may block the FTP; the full set is 124 GB.

Plan B Save it once to Drive, fall back to MIDV-500.

Hard alphabets and glare

Medium

Cyrillic, Greek and sun glare read worst.

Plan B Per-alphabet OCR, if unsure, ask a person.

Compute

Colab · Kaggle · Heidi

Cost

$0

Free and open source

All models, data and tools